Final Exam

● Graded

Student

Yixin Zhang

Total Points

36.5 / 45 pts

Question 1

Overall Grades **36.5** / 45 pts

✔ **– 0 pts** See the details on each problem.

💬 **– 8.5 pts** Point adjustment

MATH 202-001 FALL 2026 SESSION 1 FINAL EXAM
OCTOBER 19TH, 15:30–18:30

Name (Please Print)	NetID	Signature
Yixin Zhang	yz098	张艺馨

INSTRUCTION

- This is a 180-minute test.
- The total number of points is 45.
- No Cellphones.
- No Calculators.
- Do NOT write on the back of the test; if you do so, your answer(s) will NOT be graded.
- There are 9 problems.
- Show your work. To receive full credit, your answers must be neatly written, logically organized, and clearly labeled. Writing just the answer will not give you credit for that.
- The backs and extra pages are offered as scratch paper.
- GOOD LUCK.

Duke Community Standard

Duke Kunshan University (DKU) is a community comprised of individuals from diverse cultures and backgrounds. We are dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflecting upon and upholding these principles in all academic and non-academic endeavors and to protecting and promoting a culture of integrity and trust.

To uphold the DKU Community Standard

- I will hold myself to the highest standards for honesty, integrity, fairness, and responsibility in my academic and non-academic endeavors
- I will respect other cultures and embrace all forms of diversity
- I will uphold the standards if they are compromised.

Honor Pledge: I will neither give nor receive aid on this test. SIGNATURE: Yixin Zhang

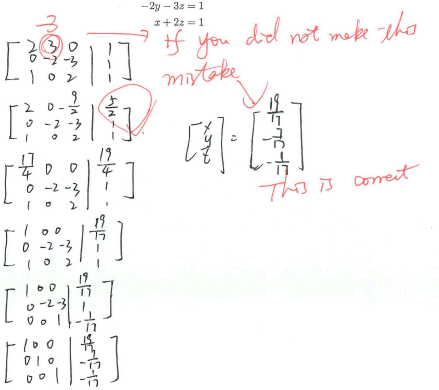
IF YOU DO NOT SIGN THE HONOR PLEDGE, YOUR FINAL EXAM WILL BE RECORDED AS 0 POINTS.

Problems	1	2	3	4	5	6	7	8	9	Total
Scores	5	5	5	5	5	5	1	0.5	5	36.5

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Problem 1. [5'] Find the general solution of the linear system:

$$2x - 3y = 1$$
$$-2y - 3z = 1$$
$$x + 2z = 1$$



Problem 2. [5] Let $A = \begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Find

(1) a basis of the column space of A,
(2) a basis of the row space of A,
(3) and a basis of the null space of A.

(sol)

$$\begin{bmatrix} 2 & 4 & 2 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ -5 & 0 & 5 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \xrightarrow{\text{sw}} \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\pm} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Pivot → 2.

(1) $\left\{ \begin{bmatrix} 2 \\ -5 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 0 \\ 1 \\ 3 \end{bmatrix} \right\}$.

(2) $\{ [2\ 4\ 2], [-5\ 0\ 5] \}$.

(3) $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.

$\begin{cases} x - z = 0 \\ y + z = 0 \end{cases}$.

$x = z \quad y = -z \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

$x - z = 1$.

Problem 3. [5] Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ be a linear transformation such that

$$T\left(\begin{bmatrix}0\\1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\1\end{bmatrix}, \qquad T\left(\begin{bmatrix}1\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}0\\2\end{bmatrix}, \qquad T\left(\begin{bmatrix}0\\0\\-1\end{bmatrix}\right) = \begin{bmatrix}2\\2\end{bmatrix}.$$

Find the matrix transformation of T, i.e., find a matrix A such that

$$T\left(\begin{bmatrix}x\\y\\z\end{bmatrix}\right) = A\begin{bmatrix}x\\y\\z\end{bmatrix}.$$

$W = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ -1 & -1 & -1\end{bmatrix}$ $V = \begin{bmatrix}1 & 0 & 2\\ 1 & 2 & 2\end{bmatrix}$

W^{-1}: $\left[\begin{array}{ccc|ccc}0 & 1 & 0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0 & 1 & 0\\ -1 & -1 & -1 & 0 & 0 & 1\end{array}\right]$

$\left[\begin{array}{ccc|ccc}0 & 1 & 0 & 1 & 0 & 0\\ 1 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 1 & -1 & -1 & -1\end{array}\right]$.

$\left[\begin{array}{ccc|ccc}1 & 0 & 0 & 0 & 1 & 0\\ 0 & 1 & 0 & 1 & 0 & 0\\ 0 & 0 & 1 & -1 & -1 & -1\end{array}\right]$

$W^{-1} = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ -1 & -1 & -1\end{bmatrix}$

~~$A = W^{-1}V = \begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ -1 & -1 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\ 1 & 2 & 2\end{bmatrix}$~~

$A = VW^{-1} = \begin{bmatrix}1 & 0 & 2\\ 1 & 2 & 2\end{bmatrix}\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 0\\ -1 & -1 & -1\end{bmatrix} = \begin{bmatrix}-2 & -1 & -2\\ 0 & -1 & -2\end{bmatrix}$.

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Problem 4. [5] Find the Jordan canonical form of

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ -2 & -2 & 1 \end{bmatrix}$$

$\det(A-\lambda I)=0$ $\quad \begin{bmatrix} 1-\lambda & 2 & 0 \\ 0 & 2-\lambda & 0 \\ -2 & -2 & 1-\lambda \end{bmatrix}$

$$(-1)^4(2-\lambda)\begin{vmatrix} 1-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} = (2-\lambda)\left[(1-\lambda)^2\right]$$

$$= (2-\lambda)(1-\lambda)^2.$$

$\lambda_1 = 2 \quad \lambda_2 = 1$

① $\lambda_1 = 2$

$$\begin{bmatrix} -1 & 2 & 0 \\ 0 & 0 & 0 \\ -2 & -2 & -1 \end{bmatrix}$$

$$\to \begin{bmatrix} -1 & 2 & 0 \\ -2 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} -1 & 2 & 0 \\ 0 & -6 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{cases} -x+2y=0 \\ -6y-z=0 \end{cases}$$

$z=t$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} -\frac{1}{3} \\ -\frac{1}{6} \\ 1 \end{bmatrix}$$

② $\lambda_2 = 1$

$$\begin{bmatrix} 0 & 2 & 0 \\ 0 & 1 & 0 \\ -2 & -2 & 0 \end{bmatrix}$$

$$\to \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\to \begin{cases} x+y=0 \\ y=0 \\ z=t \end{cases}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

⇒ Jordan Canonical form

Problem 5. [5] Diagonalize the quadratic form

$$f(x,y,z) = 3x^2 + 3y^2 + 4z^2 + 2xz - 2yz.$$

$$\begin{bmatrix} 3 & 0 & 1 \\ 0 & 3 & -1 \\ 1 & -1 & 4 \end{bmatrix}.$$

$$\begin{bmatrix} 3-\lambda & 0 & 1 \\ 0 & 3-\lambda & -1 \\ 1 & -1 & 4-\lambda \end{bmatrix}$$

$\det(A-\lambda I) = 0.$

$$(3-\lambda)\begin{vmatrix} 3-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + 1 \times \begin{vmatrix} 0 & 3-\lambda \\ 1 & -1 \end{vmatrix}$$

$$= (3-\lambda)\left[(3-\lambda)(4-\lambda) - 1\right] + 0 - (3-\lambda)$$

$$= (3-\lambda)^2(4-\lambda) - (3-\lambda) - (3-\lambda)$$

$$= (3-\lambda)\left[(3-\lambda)(4-\lambda) - 2\right].$$

$$= (3-\lambda)\left(\lambda^2 - 7\lambda + 10\right).$$

$\lambda_1 = 3$ $\quad \lambda = \dfrac{7 \pm \sqrt{49-40}}{2} = \dfrac{7 \pm 3}{2} = 5 \text{ or } 2.$

$\lambda_2 = 5 \quad \lambda_3 = 2.$

① $\lambda_1 = 3$ $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix}$

$x - y + z = 0$

$z = 0.$

$y = t$

$x = y = t$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

② $\lambda_2 = 5$ $\begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 1 & -1 & -1 \end{bmatrix} \sim \begin{bmatrix} -2 & 0 & 1 \\ 0 & -2 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$

$\begin{cases} -2x + z = 0 \\ -2y - z = 0 \\ x - y - z = 0 \end{cases}$

$x = -y - z = -2y$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

③ $\lambda_3 = 2$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = t\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\frac{1}{\sqrt{6}}\begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ $\frac{1}{\sqrt{3}}\begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}.$

$$P = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{6}}{6} & -\frac{\sqrt{3}}{3} \\ 0 & \frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{6}}{6} & \frac{\sqrt{3}}{3} \end{bmatrix}.$$

$$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

$f = X^T A X = (PU)^T A (PU)$

$= U^T (P^T A P) U.$

$= U^T D U.$

$$\begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix}\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$f = 3u^2 + 5v^2 + 2w^2.$

Problem 6. [5] Find a, b, and c such that $\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{w} = \begin{bmatrix} -3 \\ 8 \\ -8 \end{bmatrix}.$$

$$\left[\begin{array}{ccc|c} 0 & 2 & -1 & -3 \\ 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ -2 & 2 & 0 & -8 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \\ 0 & 2 & -1 & -3 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & -3 & -3 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & -1 & 1 & 8 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & 0 & 1 & 7 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

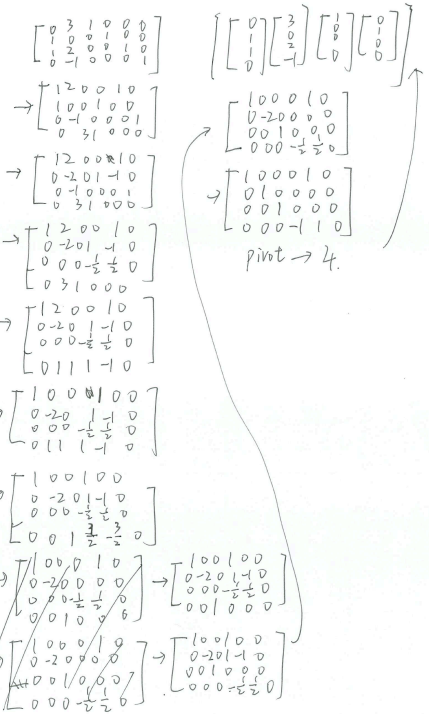
$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

$a=3 \quad b=-1 \quad c=1$

Problem 7. [5] Let V be a subspace of $\mathbb{R}^4$, defined by

$$V = span\left\{\begin{bmatrix}0\\1\\1\\0\end{bmatrix}, \begin{bmatrix}3\\0\\2\\-1\end{bmatrix}\right\}.$$

Find an orthonormal basis for V.



0.5/5

Problem 8. [5'] Let A be an $n \times n$ square matrix satisfying $A^T = -A$. Show that if n is odd, then $\det A = 0$.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \ddots & & \vdots \\ a_{n1} & \cdots & & a_{nn} \end{bmatrix}$$

~~a b c ...~~ (1*1)

→ det A can be separate into ~~ad-bc~~ the smallest ones ad-bc, and there are 0 on the diagonal of the matrix. if n is odd, there will be 0 + ~~(abc)~~ (-abc) + (abc)... except the first 0, so the remain things add together could be 0.

You can't proove by example

$\because A^T = -A$

$\therefore \begin{bmatrix} 0 & a & b & \cdots \\ -a & 0 & \ddots & \\ -b & \ddots & \ddots & \\ \vdots & & & 0 \end{bmatrix}$

antisymmetric matrix → $A^T = -A$.

$\det(A) = a_{i1}C_{i1} + \cdots + a_{in}C_{in}$

$= a_{i1}(-1)^{i+1} M_{ij} + \cdots + a_{in}(-1)^{i+n} M_{in}$.

when $i=1$ $1+n$ → even. there will be $- + \cdots - +$.

A can be ~~A=~~ $\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 3 \\ -2 & -3 & 0 \end{bmatrix}$ ⇝ $\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$ → $\begin{bmatrix} -b & -c & 0 \\ 0 & ac & c \\ 0 & a & b \end{bmatrix}$

$\det A = (1\times(-1))\begin{vmatrix} -1 & 3 \\ -2 & 0 \end{vmatrix} + 2\begin{vmatrix} -1 & 0 \\ -2 & -3 \end{vmatrix}$ → $\begin{bmatrix} -b & -c & 0 \\ 0 & ac & c \\ 0 & a & b \end{bmatrix}$.

$= -6 + (6\times 3)$

$= 0$.

Because in the matrix, there is a lease one zero one each row, ~~the pivot number~~ and the n is odd, so the pivot number must less than n.

∴ So. after Reduced the echelon form, there will be a row that all the numbers are zero.

∴ $\det A = 0$.

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Problem 9: [5] For two vectors $\vec{u}$ and $\vec{v}$, suppose they satisfy that

$$\|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\|.$$

Show that $\vec{u}$ and $\vec{v}$ are perpendicular to each other.

$\because (\vec{u}+\vec{v})^2 = \|\vec{u}\|^2 + 2\vec{u}\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2$

$(\vec{u}-\vec{v})^2 = \|\vec{u}\|^2 - 2\vec{u}\vec{v} + \|\vec{v}\|^2 = \|\vec{u}\|^2 - 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2$

$\because \|\vec{u}+\vec{v}\| = \sqrt{(\vec{u}+\vec{v})^2}$

$\|\vec{u}-\vec{v}\| = \sqrt{(\vec{u}-\vec{v})^2}$

$\therefore \|\vec{u}+\vec{v}\| = \|\vec{u}-\vec{v}\| \Rightarrow \sqrt{\|\vec{u}\|^2 + 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2} = \sqrt{\|\vec{u}\|^2 - 2\|\vec{u}\|\|\vec{v}\|\cos\theta + \|\vec{v}\|^2}$

$\Rightarrow +2\|\vec{u}\|\|\vec{v}\|\cos\theta = -2\|\vec{u}\|\|\vec{v}\|\cos\theta$

$\therefore \cos\theta = 0$

$\therefore \vec{u}$ and $\vec{v}$ are perpendicular to each other.